# Rebecca Ann Jackson

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## EDUCATION

**Villanova University – Villanova, PA**

– M.S. in Computer Science | GPA: 4.00 — Aug 2025 – May 2027

– B.S. in Computer Science & B.S. in Mathematics | GPA: 3.99 — Aug 2023 – Jul 2026

## WORK EXPERIENCE

**Graduate Assistant | Villanova University** — Aug 2026 – Present

– Selected for a fully funded graduate assistantship with the Computer Science department, contributing 20 hours per week in a teaching or research capacity while being a full-time student

**Business Insights & Analytics Intern | Wawa Corporate Headquarters, Media, PA** — May – Aug 2026

– Worked within the Finance department under Planning & Analysis, supporting data-driven business decision-making

– Queried and analyzed millions of rows of data using SQL in Snowflake; created dashboards and visualizations in Excel and Tableau

– Built sales forecasting models using machine learning to support financial planning and business strategy

**Undergraduate Teaching Assistant | Villanova University** — Jan 2025 – May 2026

– Served as TA for Competitive Programming and Data Structures & Algorithms courses

– Provided detailed, actionable feedback on assignments for 30+ students and hosted weekly office hours

**Undergraduate Research Assistant | Villanova University** — Jan 2024 – Aug 2026

– Contributed to 4 projects through Python data analysis, API integration, and web development

– Co-authored 2 peer-reviewed publications: Mathematics Magazine (Jul 2025) and NordiCHI 2024 Proceedings

## TECHNICAL SKILLS

– **Languages:** Python, SQL, R, Java, C, C++, SML, JavaScript, HTML, CSS

– **Frameworks and Tools:** Pandas, NumPy, Scikit-Learn, PyTorch, Jupyter, Tableau, Excel, Snowflake, Git, GitHub, React

– **Areas:** Machine Learning, Data Visualization and Analysis, Data Cleaning, Data Reporting, Web Development

## PROJECTS

**Carbon Calculus – React, Recharts, Vite, JavaScript, HTML/CSS, Git** — Jun 2025 – Present

– Built an interactive $CO_2$ emissions modeling tool featuring lifecycle cost analysis, IRR calculations, and multi-scenario comparison with dynamic data visualizations

– First undergraduate project funded by Villanova's Strategic Initiative for Climate, Justice, and Sustainability

**American Sign Language Classifier – Python, PyTorch, Scikit-Learn, Git** — Aug – Dec 2025

– Trained a convolutional neural network to classify 36 ASL numbers and letters with approximately 95% accuracy

– Documented end-to-end reproducibility in a public GitHub repository for full model transparency

## RESEARCH PUBLICATIONS

**#Haul: Analyzing YouTube Discourse on Over-Consumption & Sustainability** — NordiCHI 2024

– Implemented Python scripts and API integration to collect and analyze 1,000 YouTube videos, applying data modeling, text analysis, and visualization techniques

**Non-Recursive Solution of the Tower of Hanoi** — Mathematics Magazine, Jul 2025

– Collaborated with academics to derive the first non-recursive equation for the Tower of Hanoi; built an interactive JavaScript demo site

## HONOR SOCIETIES

**Pi Mu Epsilon** – National mathematics honor society — May 2026

**Upsilon Pi Epsilon** – International honor society for computing and information disciplines — Mar 2026

**Phi Beta Kappa** – Oldest academic honor society; selected for excellence in the liberal arts and sciences — May 2025

## LEADERSHIP AND ACTIVITIES

**President, Girls Who Code Villanova** — Apr 2025 – Present

– Lead a community of 50+ women through educational, social, and technical meetings and events

**Founder, Aspirations in Computing Philadelphia Affiliate** — Jan 2025 – May 2026

– Organize an annual NCWIT banquet for 20+ Philadelphia winners of the Aspirations in Computing Award

**President, Villanova Programming Team** — Sep 2024 – Present

– Lead weekly competitive programming practice sessions for 130+ members in Python, C++, and Java

– Led team to top finishes at ICPC regionals: 1st place at competition site in Division 2 (2024) and 2nd place in Division 1 (2025)